

Sensitivity of tree species demography to climate and competition across their range

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31 **Abstract**

32 Theory predicts that demographic performance should peak at the core of species ranges
33 and decrease toward their limits. Yet, empirical correlations between population growth
34 rate and species distribution remain weak for most tree species. Part of the problem
35 may arise from the difficulty of integrating multiple demographic processes across the
36 complex life cycle of a forest, and from the significant variability among individuals and
37 locations. It remains unclear if the mismatch between performance and distribution arises
38 from modelling limitations or if climate is simply a poor predictor of species performance
39 across distributions. Here, rather than asking whether demographic performance correlates
40 with species distributions, we ask how climate and competition jointly shape population
41 growth rate for 31 tree species across eastern North America. By combining flexible
42 nonlinear hierarchical models for growth, survival, and recruitment with explicit uncertainty
43 propagation, we use Integral Projection Models to address key gaps in previous studies.
44 Perturbation analyses revealed that population growth rate was consistently more sensitive
45 to mean annual temperature than to conspecific or heterospecific competition across all
46 species. We further examined how sensitivities to climate and competition varied across
47 species' thermal ranges. The dominance of climate over competition increased toward both
48 cold and hot range limits, while sensitivity to competition generally declined from cold to
49 hot limits. Notably, these patterns emerged along the continental thermal gradient shared
50 across species rather than within each species' individual range, suggesting that range-
51 edge demographic responses may arise as a community-level phenomenon. Across species,

the largest source of variability remained the local plot conditions captured by random effects, likely reflecting differences in soil conditions, drainage, and disturbance history. Together, these results may provide a mechanistic pathway underlying the performance declines predicted by range-limit theories, and offer a basis for understanding how forest populations and communities may reorganize in response to ongoing climate change and shifting disturbance regimes.

Keywords: Integral Projection Models, Tree demography, Species distribution, Range limits, Perturbation analysis

1 Introduction

Mechanistic and process-based models (Evans et al. 2016) have been proposed to improve predictions of species distribution in response to climate change. Among these, demographic range models predict species distributions from individual performance governed by growth, survival, and recruitment rates (Pagel and Schurr 2012). This framework assumes that population growth rate (λ), determined by demographic rates, varies across environmental gradients, with species range limits occurring where λ is positive (Maguire Jr 1973, Holt 2009, Pang et al. 2024). Approaching species distributions from a demographic perspective allows a more complete representation of how environmental variation and species interactions shape forest dynamics (Schurr et al. 2012, Svenning et al. 2014). Several studies have attempted to predict tree species distributions from demographic performance. The simplest implementations rely on environment-dependent demographic rates to estimate λ (e.g. Merow et al. 2014, Csergő et al. 2017). However, competition is a fundamental determinant of demographic rates (Clark et al. 2011, Luo and Chen 2011, Zhang et al. 2015) and population performance (Scherrer et al. 2020, Le Squin et al. 2021) in forest ecosystems. Competition is therefore a necessary addition given the projected risk of community reshuffling under global change (Alexander et al. 2016). This more complete, realized expression of the niche (Hutchinson 1957) may help explain why North American forest trees often fail to occupy their full climatically suitable ranges (Boucher-Lalonde et al. 2012, Talluto et al. 2017).

An increasing body of evidence challenges theoretical expectations by reporting weak correlations

79 between tree demographic performance and species distributions (McGill 2012, Csergő et al. 2017,
80 Bohnert and Diez 2020, Le Squin et al. 2021, Midolo et al. 2021, Guyennon et al. 2023, Thuiller et
81 al. 2014). The methodological approaches used to quantify demographic performance are however
82 rarely challenged, difficult to compare and suffer some limitations. Some analyses infer performance
83 from a single proxy, such as radial growth (e.g. McGill 2012, Bohnert and Diez 2020). Even when
84 demographic rates are combined within population models, key components such as recruitment are
85 often excluded due to data limitations (Kunstler et al. 2021, Le Squin et al. 2021). Furthermore,
86 density-dependence is sometimes ignored altogether (Csergő et al. 2017, Ohse et al. 2023), and when
87 included, studies rarely distinguish between conspecific and heterospecific competition (Bohner and Diez
88 2020, Le Squin et al. 2021). Finally, despite increasing recognition of the importance of propagating
89 model and data uncertainty (Milner-Gulland and Shea 2017, Malchow and Hartig 2024), most studies
90 evaluate performance under average environmental conditions and rely on pointwise estimates, missing
91 the substantial variability inherent to demographic processes.

92 Rather than asking whether demographic performance correlates with species distributions, a more
93 fruitful question may be how climate and competition jointly shape demographic performance. Despite
94 substantial progress, we still lack a full partitioning of the sensitivity of forest dynamics to local and
95 regional drivers (Ohse et al. 2023). For instance, Clark et al. (2011) found that annual growth was more
96 sensitive to competition, whereas fecundity responded more strongly to climate. In contrast, Copenhaver-
97 Parry and Cannon (2016) reported growth to be more sensitive to climate than to competition. Although
98 these studies provide important insights into how forest trees may respond to climate change and
99 management, they focus on individual demographic components rather than their integrated effects
100 on population growth. The distinction matters if sensitivity to climate and competition varies across
101 life-history stages (Russell et al. 2012, Ettinger and HilleRisLambers 2013). The sensitivity of λ to
102 climate and competition may also depend on a species' position within its range. For instance, climate
103 may exert stronger control under abiotic stress, whereas competition may dominate under more benign
104 conditions (Louthan et al. 2015). Yet such range-dependent evaluations of demographic sensitivity
105 remain largely unexplored for forest trees (Ohse et al. 2023).

106 Here, we evaluate how climate and competition jointly shape the demography and population growth
107 rate of the 31 most abundant forest tree species across eastern North America (Figure 1). Using

108 forest inventories spanning 26–53° latitude across the United States and Canada, we capture the
109 broad geographic ranges of these species. We model growth, survival, and recruitment as functions of
110 temperature, precipitation, and conspecific and heterospecific basal area (as proxies for competition)
111 using Bayesian hierarchical models, with explicit uncertainty propagation. We then integrate these
112 demographic models into size-structured Integral Projection Models (IPMs) to estimate population
113 growth rate (λ) under specified climate and competition conditions.

114 Our primary goal is to quantify how sensitive λ is to climate and competition across species' ranges. First,
115 we estimate sensitivities at each plot under observed environmental and competitive conditions using
116 perturbation analysis (Caswell 2000). This approach ensures that sensitivities reflect the environmental
117 conditions actually occupied by the populations. Second, building on previous evidence that North
118 American trees have shown limited cold-edge expansion and hot-edge contraction (Talluto et al. 2017),
119 we test whether demographic sensitivity varies systematically with species' thermal range position. By
120 quantifying how climate and competition jointly regulate population growth, this framework provides a
121 mechanistic basis for predicting tree responses to climate change, management, and conservation.

122 [Figure 1 about here.]

123 2 Methods

124 We fitted species-specific growth, survival, and recruitment models using forest inventory data and
125 integrated them into an Integral Projection Models (IPMs) to estimate population growth rate (λ).
126 We then used perturbation analyses to quantify the sensitivity of λ to climate and competition across
127 species-specific thermal range positions.

128 2.1 Forest inventory and climate data

129 We used two open forest inventory datasets from eastern North America: the Forest Inventory and
130 Analysis (FIA) dataset in the United States (O'Connell et al. 2007) and the Forest Inventory of Québec
131 (Ministère des Ressources Naturelles 2016). At the plot level, we focused on plots sampled at least
132 twice and excluded those that had undergone harvesting in order to concentrate solely on natural forest
133 dynamics. Specifically for the FIA dataset, we selected surveys conducted using the modern standardized

methodology implemented since 1999. After applying these filters, the final dataset encompassed nearly 26,000 plots spanning a latitudinal range from 26° to 53° (Figure S7). Each plot was measured between 1970 and 2021, with observation frequencies ranging from 2 to 7 measurements and an average of 3 measurements per plot. The time intervals between measurements varied from 1 to 40 years, with a median interval of 7 years (Figure S7).

These datasets provide individual-level information on diameter at breast height (DBH) and survival status (alive or dead) for more than 200 species. From this pool, we selected the 31 most abundant species (Table S1), comprising 9 conifer species and 22 hardwood species. We ensured broad coverage across the shade tolerance gradient, including three very intolerant species, nine intolerant species, eight intermediate species, eight tolerant species, and five very tolerant species (Burns et al. 1990).

We obtained 19 bioclimatic variables at a spatial resolution of 10 km² (300 arcsec), covering the period from 1970 to 2018. These climate variables were modeled using the ANUSPLIN interpolation method (McKenney et al. 2011). Using the geographic coordinates of each plot, we extracted mean annual temperature (MAT) and mean annual precipitation (MAP). When plots did not fall within a valid climate pixel, values were interpolated using the eight neighboring cells. Because measurements occurred across multiple years within census intervals, we calculated both the mean and standard deviation of MAT and MAP across the years included in each time interval.

2.2 Model

We developed an Integral Projection Model (IPM) for 31 tree species, with separated sub-models for each demographic rate. An IPM is a mathematical framework used to represent the dynamics of structured populations and communities. Unlike traditional matrix population models, IPMs represent individual traits as continuous variables in discrete time (Easterling et al. 2000). This formulation enables a detailed representation of trait transitions between time steps, which is particularly relevant for trees due to the considerable variability in demographic rates (Kohyama 1992, Clark et al. 2011, Le Squin et al. 2021). Specifically, the IPM predicts the transition of a distribution of individual traits from time t to time $t + \Delta t$, where Δt represents the number of years between observations:

$$n(z', t + \Delta t) = \int_L^U K(z', z, \theta) n(z, t) dz \quad (1)$$

160 The continuous trait z represents DBH, bounded between lower (L) and upper (U) limits, and $n(z, t)$
 161 describes the DBH distribution at time t . Transitions from $n(z, t)$ to $n(z', t + \Delta t)$ are governed by the
 162 kernel K and species-specific parameters θ . The kernel K , a continuous analogue of the projection
 163 matrix in structured population models, is composed of three demographic components:

$$K(z', z, \theta) = [Growth(z', z, \theta) \times Survival(z, \theta)] + Recruitment(z, \theta) \quad (2)$$

164 The growth function describes changes in DBH, the survival function determines the probability of
 165 remaining alive over the next interval, and the recruitment function describes the influx of new individuals.
 166 These three models are estimated independently and do not share parameters. Below, we first describe
 167 the baseline versions of these models, followed by the inclusion of climate and competition covariates.

168 2.2.1 Demographic rates

169 For the growth model, the DBH of an individual at time $t + \Delta t$ after growing from time t follows the
 170 distribution:

$$dbh_{i,t+\Delta t} \sim N(\mu_{i,t+\Delta t}, \sigma) \quad (3)$$

171 We used the von Bertalanffy growth equation to describe annual DBH growth for individual i (Von
 172 Bertalanffy 1957). The expected size at time $t + \Delta t$, given initial size $dbh_{i,t}$, is:

$$\mu_{i,t+\Delta t} = dbh_{i,t} e^{-\Gamma \Delta t} + \zeta_{\infty} (1 - e^{-\Gamma \Delta t}) \quad (4)$$

173 Here, Δt represents the time interval between measurements, Γ is a dimensionless growth-rate parameter,
 174 and ζ_{∞} represents the asymptotic size at which growth approaches zero. This formulation assumes that

175 growth declines exponentially with size, converging to zero as size approaches ζ_∞ . This assumption is
 176 particularly valuable in the context of an IPM, as it prevents eviction beyond the bounds of the IPM
 177 domain $[L, U]$.

178 For survival, a mortality event (M) for individual i during the interval between t and $t + \Delta t$ is modeled
 179 as:

$$M_i \sim \text{Bernoulli}(p_i) \quad (5)$$

180 Here, M_i denotes survival status and p_i represents mortality probability. Mortality probability is derived
 181 from the annual survival rate (ψ):

$$p_i = 1 - \psi^{\Delta t} \quad (6)$$

182 Thus, survival probability increases with longevity parameter ψ , while longer intervals Δt increase
 183 mortality risk exponentially.

184 For recruitment, we combined U.S. and Québec inventory data to capture a broader climatic range.
 185 However, these inventories differ in protocols for recording seedlings and saplings, including size thresholds.
 186 We therefore defined recruitment as the ingrowth of individuals into the adult population ($\text{DBH} \geq 12.7$
 187 cm).

188 Recruitment rate (I) integrates fecundity, dispersal, growth, and survival up to the threshold size.
 189 Because ingrowth depends on census interval length, we defined two parameters: ϕ , the annual ingrowth
 190 rate per square meter, and ρ , the annual survival probability of recruits:

$$I \sim \text{Poisson}\left(\phi \times A \times \frac{1 - \rho^{\Delta t}}{1 - \rho}\right) \quad (7)$$

191 Here, A represents plot area (m^2). Individuals are assumed to enter annually at rate ϕ , while their

192 survival (ρ) until the subsequent measurement declines with time. Note that ρ (recruit survival) is
 193 distinct from ψ (adult annual survival in Equation 6); the two parameters are estimated independently.
 194 ρ is estimated from the data of individuals arriving in the population. Once individuals are recruited
 195 into the population, their initial size z_I is modeled as:

$$z_I \sim TNormal(\Omega + \gamma_{rec}\Delta t, \sigma, L_{rec}, \infty) \quad (8)$$

196 This truncated normal distribution has lower bound $L_{rec} = 12.7$ cm and no upper bound. The mean
 197 recruit size at observation increases with census interval length at a rate γ_{rec} .

198 2.2.2 Covariates

199 Plot-level random effects were included in all demographic components to account for shared environ-
 200 mental variation among individuals within plots.

$$\begin{aligned} \alpha_j &\sim N(0, \sigma) \\ \eta_j &= \bar{\eta} + \alpha_j \end{aligned} \quad (9)$$

201 Here, η represents parameters Γ , ψ , or ϕ depending on the demographic component, and σ the variance
 202 among all plots.

203 We assumed that competition for light is the primary competitive factor driving forest dynamics (Pacala
 204 et al. 1996). We therefore considered that each individual was affected only by larger neighbors. We
 205 quantified competition for light for a focal individual in a given plot by summing the basal area of
 206 all individuals larger than the focal one, herein BAL. We further split BAL into the total density of
 207 conspecific and heterospecific individuals.

208 Competition effects were modeled on parameters Γ , ψ , and ρ :

$$\eta + \beta(BAL_{cons} + \theta \times BAL_{het}) \quad (10)$$

Parameter θ controls the relative strength of heterospecific competition. Values $\theta < 1$ indicate stronger conspecific competition, $\theta > 1$ indicates stronger heterospecific effects, and when $\theta = 1$, there is no distinction between them. Note that β is also unbounded, allowing it to converge towards negative (indicating competition) or positive (indicating facilitation) values. For recruitment survival (ρ), θ was fixed at 1 due to convergence issues. Recruitment rate ϕ also included conspecific density-dependence, where it decreases with BAL_{cons} as a positive effect of seed source up to reach the optimal density of recruitment, δ , and then decreases with more conspecific density due to competition at a rate proportional to σ :

$$\phi + \left(\frac{BAL_{cons} - \delta}{\sigma} \right)^2 \quad (11)$$

For climate, we used MAT and MAP as climate predictors because of their widespread use in species distribution modeling and demonstrated relevance to model demography of these species (Le Squin et al. 2021). Each demographic parameter followed a unimodal climate response determined by an optimal climate condition (ξ) and a climate breadth parameter (σ):

$$\eta + \left(\frac{MAT - \xi_{MAT}}{\sigma_{MAT}} \right)^2 + \left(\frac{MAP - \xi_{MAP}}{\sigma_{MAP}} \right)^2 \quad (12)$$

The climate breadth parameter (σ) influences the strength of the specific climate variable's effect on each demographic component. This unimodal function is flexible, assuming various shapes, such as bell, quasi-linear, or flat shapes. However, this flexibility introduces the possibility of parameter degeneracy or redundancy, where different combinations of parameter values yield similar outcomes. To address this issue, we constrained the optimal climate condition parameter (ξ) within the observed climate range for the species, assuming that the optimal climate condition falls within our observed data range.

2.2.3 Model fit and validation

We fitted growth, survival, and recruitment models separately for each species using Hamiltonian Monte Carlo (HMC) implemented in Stan (version 2.30.1 Team and Others 2022) through the `cmdstanr` R interface (version 0.5.3 Gabry et al. 2023). Weakly informative priors were used for all parameters and their full specifications are documented in Table S2. Each model used four chains with 2000 warm-up iterations and 2000 sampling iterations, yielding 8000 posterior samples. To reduce storage, only the final 1000 samples from each chain were retained, resulting in 4000 posterior samples. Models were constructed incrementally, beginning with intercept-only models and progressively adding random effects, competition, and climate covariates. Our goal was not maximal predictive accuracy but mechanistic inference (Tredennick et al. 2021). We focused on assessing the relative effects of climate and competition while controlling for other influential factors. Therefore, our modeling approach is guided by biological mechanisms, which tend to provide more robust extrapolation (Briscoe et al. 2019) rather than solely by statistical metrics. We also verified that increasing model complexity did not reduce predictive performance using mean squared error (MSE), pseudo- R^2 (Gelman et al. 2019), and Leave-One-Out Cross-Validation (LOO-CV). Additional diagnostics are provided in Supplementary Material 1.

The IPM kernel K was constructed following Equation 2 and discretized using the midpoint rule (Ellner et al. 2016). Kernel discretization used 0.1 cm size bins, consistent with previous recommendations for trees (Zuidema et al. 2010). The asymptotic population growth rate (λ) was computed as the leading eigenvalue of the discretized kernel matrix under specified climate and competition conditions. All model code is available in the `TreesDemography` repository. The IPM implementation is packaged in `forestIPM`, and sensitivity analysis code is available in the `simulations/covariates_perturbation` directory.

2.3 Perturbation analysis

Sensitivity was defined as the partial derivative of λ with respect to a covariate X , representing either competition (conspecific or heterospecific density-dependence) or climate (temperature or precipitation). We quantified sensitivity by slightly increasing each covariate value X_j to X'_j and computing the resulting change in λ using a finite-difference approximation (right-hand side of Equation 13):

$$\left. \frac{\partial \lambda_{ij}}{\partial X_j} \right|_{K_{ij}} \approx \frac{\Delta \lambda_{ij}}{\Delta X_j} = \frac{|f(X'_j) - f(X_j)|}{X'_j - X_j} \quad (13)$$

254 Sensitivity was evaluated separately for each species i and plot-year j , conditional on the local climate
 255 and competition conditions as well as the corresponding kernel parameters K_{ij} . We set the perturbation
 256 magnitude to a 1% increase on the normalized scale of each covariate. For instance, a 1% increase
 257 corresponds to approximately 0.3°C for mean annual temperature (MAT) and 26 mm for mean annual
 258 precipitation (MAP). Because competition metrics were computed at the individual level, perturbations
 259 were applied to each individual prior to calculating plot-level basal area, where a 1% increase corresponds
 260 approximately to an increase of 1.2 cm in DBH. As we were interested in the magnitude of the response,
 261 absolute differences were used, resulting in sensitivity values ranging from 0 to ∞ , where lower values
 262 indicate weaker sensitivity of λ to the given covariate. Specifically, sensitivity (S) to competition or
 263 climate for species i at plot-year j was defined as:

$$\begin{aligned}
 S_{comp,ij} &= \frac{\partial \lambda_{ij}}{\partial BAL_{cons,i}} + \frac{\partial \lambda_{ij}}{\partial BAL_{het,i}} \\
 S_{clim,ij} &= \frac{\partial \lambda_{ij}}{\partial MAT_i} + \frac{\partial \lambda_{ij}}{\partial MAP_i}
 \end{aligned} \quad (14)$$

264 When averaging $S_{X,i}$ across plot-years j , this metric represents the sensitivity of λ_i to covariate X ,
 265 conditional on the observed probability distribution of that covariate. This yields a measure of realized
 266 sensitivity, integrating both demographic responses and the environmental variability experienced across
 267 the species' range. To evaluate range-dependent sensitivity, plots were categorized into cold, center, or
 268 hot conditions along the MAT gradient for each species. Plots were classified as cold (or hot) if their
 269 average MAT fell below the 10th percentile (or above the 90th percentile) of the species-specific MAT
 270 distribution, with intermediate plots classified as center plots. Sensitivity within each range position was
 271 calculated as the average sensitivity across plots belonging to that category. Because this classification is
 272 based on observed MAT distributions, range categories are conditional on the environmental conditions
 273 experienced by each species.

3 Results

3.1 Model validation

All species-specific demographic components demonstrated good convergence ($\hat{R} < 1.05$) with few to no divergent iterations. Model comparison using LOO-CV consistently favored the full model, featuring plot-level random effects, competition, and climate covariates, over other competing models for all three demographic rates (Supplementary Material 1). The magnitude of LOO-CV differences indicated that the growth model benefited most from the inclusion of covariates, followed by recruitment and survival. We further evaluated biological realism by comparing model-derived parameters with known species trait groups, including growth rate class, maximum observed size, maximum observed age, shade tolerance, and seed mass (Burns et al. 1990, Díaz et al. 2022).

The growth model intercept included two parameters: one determining asymptotic size (ζ_∞) and another describing annual growth rate (Γ). Predicted asymptotic size, which can be interpreted as the maximum predicted size of the species, correlated well with maximum observed species size reported in the literature ($R^2 = 0.31$, Figure 2). Γ values aligned with established fast-, moderate-, and slow-growing trait classes (Figure S8). In the survival model, expected longevity (L) was derived from annual survival probability (ψ) using the relationship $L = e^\psi$. Estimated longevity showed strong agreement with maximum observed age from the literature ($R^2 = 0.59$, Figure 2). For recruitment, the log of annual ingrowth rate (ϕ) declined linearly with seed mass (Figure S9), consistent with the seed mass–growth rate trade-off (Reich et al. 1998). The annual survival probability of ingrowth (ρ) decreased with increasing shade intolerance (Figure S10).

[Figure 2 about here.]

Both conspecific and heterospecific competition effects in growth and survival models increased with shade intolerance (Figure 3). Across nearly all species, conspecific competition effects were stronger than heterospecific effects. Only two species in the growth model and three in the survival model showed stronger heterospecific than conspecific competition. *Fagus grandifolia* and *Thuja occidentalis* were exceptions, exhibiting positive density-dependence in survival. In the recruitment model, total stand density effects increased with shade intolerance (Figure S11).

[Figure 3 about here.]

The distribution of optimal MAT (ξ_{MAT}) and MAP (ξ_{MAP}) indicated that optimal conditions for growth, survival, and recruitment were rarely located at the center of species' ranges (Figures S12 and S13). Many species exhibited demographic compensation, with opposing responses among demographic rates to environmental variation (Villellas et al. 2015). Climate breadth (σ) determined how narrowly or broadly performance varied across MAT and MAP. Across species, climate breadth increased with geographic range size, indicating that widely distributed species tended to have broader niche breadths (Figure S14). An exception was survival breadth along MAT, which showed only a weak relationship.

3.2 λ sensitivity to climate and competition

We conducted perturbation analyses to quantify the relative contribution of each covariate to variation in population growth rate (λ). Figure 4 summarizes the sensitivity of λ to conspecific competition, heterospecific competition, temperature, and precipitation, averaged across all plot-year observations. Across species, λ was most sensitive to temperature, followed by conspecific and heterospecific competition; sensitivity to precipitation was substantially weaker, and this pattern was consistent across species.

[Figure 4 about here.]

To evaluate range-dependent responses, plots were divided into cold and hot regions based on species-specific MAT distributions (Figure 5). Most species occurring in colder climates exhibited decreasing climate sensitivity from the cold to the hot limit of their range. In contrast, species centered in warmer climates generally showed greater climate sensitivity at the hot limit than at the cold limit. Sensitivity to competition was typically higher at the cold limit than at the hot limit across most species, regardless of overall distribution. This elevated sensitivity to competition at the cold limit was particularly pronounced among boreal species.

[Figure 5 about here.]

Including sensitivity estimates from the central portion of species ranges further clarified patterns across thermal gradients (Figure S15). Most species displayed approximately linear changes in sensitivity to

327 both climate and competition from cold to hot range limits. For climate sensitivity, three of the four
328 species exhibiting concave relationships, where sensitivity was highest at both range edges, were among
329 those with the largest geographic ranges. In contrast, four species displayed convex relationships in
330 competition sensitivity, with peak sensitivity occurring near the range center. These species also showed
331 the highest overall competition sensitivity and were predominantly distributed in colder climates.

332 Finally, we evaluated how the relative importance of climate versus competition varied across species'
333 thermal positions (Figure 6). For most species, λ remained more sensitive to climate than to competition
334 across cold, center, and hot range positions. The relative importance of climate increased toward both
335 cold and hot limits along the MAT gradient. Populations near the thermal extremes of the continental
336 climate gradient were therefore more strongly climate-sensitive than those located near the center of the
337 continental climate space. However, the mechanisms underlying this increase differed between range
338 limits (Figure S16). At the cold limit, sensitivities to both climate and competition increased, but the
339 proportional increase was greater for climate. In contrast, at the hot limit, the increasing dominance of
340 climate sensitivity primarily resulted from declining sensitivity to competition.

341 [Figure 6 about here.]

342 4 Discussion

343 We developed an Integral Projection Model for 31 tree species to evaluate the sensitivity of λ to climate
344 and competition. Our framework extends previous analyses of tree species performance by explicitly
345 incorporating climate and competition effects into the recruitment model, distinguishing conspecific
346 from heterospecific competition, and propagating uncertainty across all modeling levels. The modular
347 structure also allows straightforward expansion to the more than 200 species available in the dataset, as
348 well as additional environmental covariates affecting demographic rates.

349 Adding climate and competition covariates consistently improved predictive performance across all
350 demographic components relative to models containing only random effects. Local plot-level conditions,
351 however, remained the strongest predictors of demographic variation, so we evaluated climate and
352 competition sensitivities while accounting for this plot-level variability. Across species and their

geographic ranges, λ was most sensitive to temperature and to conspecific basal area of larger individuals. These sensitivities varied across species' range positions, with climate exerting relatively stronger influence than competition at both cold and hot range limits. Together, these findings contribute to a more mechanistic understanding of how tree species may respond to novel environmental conditions associated with climate change, forest management, and conservation.

4.1 Fit of demographic components

Our model demonstrated strong biological consistency by reproducing well-established relationships among demographic traits. Growth and survival intercepts were positively correlated with maximum observed size and longevity, respectively (Burns et al. 1990), while the recruitment intercept showed clear alignment with seed mass (Díaz et al. 2022). The model also reproduced the fast-slow life-history continuum (Salguero-Gómez et al. 2016), reflected in a negative relationship between growth and survival rates and a positive relationship between growth and recruitment rates (Figure S17). Competition effects were consistent with ecological expectations. The model captured the negative relationship between density-dependence and shade tolerance, as well as the general pattern of stronger responses to conspecific competition than to heterospecific competition. This asymmetry is considered fundamental for species coexistence and biodiversity maintenance (Chesson 2000). Conspecific density-dependence was also stronger for fast-growing species than for slow-growing ones (Figure S18), consistent with observations from tropical forests (Zhu et al. 2018). Validation of climate-related parameters remains challenging due to limited empirical estimates of species-specific climatic optima. Nevertheless, our results are aligned with previous findings documenting demographic compensation among forest trees (Bohner and Diez 2020, Yang et al. 2022). The estimated climatic breadth of species also correlated with geographic range size (Figure S14), suggesting that the model captures ecologically meaningful variation beyond explicitly modeled predictors.

Most of the variation in λ was associated with local plot conditions captured by random effects, consistent with previous studies (Vanderwel et al. 2016, Le Squin et al. 2021, Itter and Finley 2024). This result indicates that demographic variability is influenced by drivers not explicitly included in our models. At local scales, for instance, soil nitrogen availability (Ibáñez et al. 2018) and mixed mycorrhizal associations (Luo et al. 2023) can enhance tree growth rates. At broader spatial scales, disturbance

381 regimes such as wildfire and insect outbreaks strongly shape forest dynamics and structure (Franklin et
382 al. 2002), causing synchronized mortality events and altering species composition across landscapes.
383 Structured spatial variation in mortality among *Pinus* species across the United States has similarly
384 been linked primarily to local disturbance regimes rather than broad climatic gradients (Bauman et al.
385 2025). Although our analysis focused on climate and competition, other covariates may exert stronger
386 influences on demographic variation. For instance, tree growth models incorporating extreme climatic
387 events often show improved predictive performance relative to those using mean climatic conditions
388 (Sanginés de Cárcer et al. 2017). Drought extremes, rather than mean precipitation, have also been
389 identified as strong predictors of fecundity following temperature effects (Clark et al. 2011).

390 4.2 λ sensitivity to climate and competition

391 We found across all species that the sensitivity of λ was highest for temperature, followed by conspecific
392 competition. Previous studies assessing the relative importance of climate and competition on tree
393 performance have reported mixed results. Some studies emphasize stronger competition effects on
394 growth (Gómez-Aparicio et al. 2011, Le Squin et al. 2021), whereas others report stronger climate
395 effects (Copenhaver-Parry and Cannon 2016). The relative importance of climate and competition
396 also varies among demographic components, with growth typically more sensitive to competition and
397 fecundity more sensitive to climate (Clark et al. 2011). For instance, several dominant tree species
398 in Poland exhibited high climate sensitivity, with declining fecundity under warming temperatures
399 (Foest et al. 2025). These contrasting findings may arise because many studies assess demographic
400 components separately rather than evaluating their integrated effects on population growth rate. This
401 matters because λ does not respond equally to all demographic processes. Our additional analyses
402 revealed that λ was most sensitive to recruitment, followed by survival, with a comparatively smaller
403 contribution from growth (Supplementary Material 3). Because recruitment tends to be more sensitive
404 to temperature (Clark et al. 2011, Foest et al. 2025), the high elasticity of λ to recruitment may explain
405 the dominant role of climate sensitivity observed in our results.

406 Demographic sensitivity to climate and competition varied across species' geographic ranges. Because
407 we modeled climate effects as a unimodal function centered at an optimal value, lower sensitivity values
408 in our framework indicate conditions close to the climatic optimum, whereas higher values indicate

409 deviation from optimal conditions. Overall, sensitivity of λ to climate, driven primarily by MAT, was
410 greatest at both cold and hot range limits. This result suggests that species occupying colder portions
411 of the continental thermal gradient have their demographic optimum toward the warmer parts of their
412 ranges, whereas species occupying warmer portions of the continental gradient have their optimum
413 toward the cooler parts. Because multiple species occupy overlapping portions of the same thermal
414 gradient, increasing climatic sensitivity near thermal extremes may represent a shared demographic
415 response to environmental stress rather than an independent property of each species' geographic range
416 (Körner 2021). This interpretation shifts the focus from species-level range position to community-level
417 responses to climate, highlighting potential common physiological constraints experienced across species.

418 The demographic mechanisms underlying increased climate sensitivity differed between range limits.
419 Recruitment and growth processes primarily drove sensitivity at cold range limits, whereas survival
420 and recruitment dominated at hot range limits, particularly under high competition conditions (Figure
421 S22). Previous studies have documented climate-constrained growth at cold range limits for both North
422 American (Ettinger and HilleRisLambers 2013) and European trees (Kunstler et al. 2021). Reduced
423 survival at hot range limits has been observed in European forests (Kunstler et al. 2021), although this
424 observation has not been consistently detected in eastern North America (Purves 2009).

425 Sensitivity of λ to competition increased approximately linearly toward colder climates for most species.
426 Due to nonlinear relationships between demographic performance and competition, sensitivity declines
427 as stand density increases, following a negative exponential pattern. The reduction in competition
428 sensitivity toward hot range limits likely reflects higher overall stand density in those regions. Biotic
429 interactions are often thought to dominate at hot range limits (Paquette and Hargreaves 2021). However,
430 when focusing exclusively on growth responses, previous studies have reported relatively constant
431 competition effects across climatic gradients in both North American (Ettinger and HilleRisLambers
432 2013) and European forests (Kunstler et al. 2011).

433 4.3 Limitations and future perspectives

434 Structured population models such as IPMs are essential for capturing ontogenetic variation in population
435 dynamics. While our growth model explicitly incorporates individual size, survival and recruitment

models were specified without direct size dependence. Attempts to include size-dependent mortality using the widely assumed U-shaped function (Lines et al. 2010) did not improve model performance relative to simpler random-effects formulations (Figure S6). Although mortality often increases with size (Luo and Chen 2011, Hember et al. 2017), its significance appears to manifest only when interacting with climate and competition (Le Squin et al. 2021). The challenge in estimating size-dependent survival likely stems from limited observations of small individuals ($\text{dbh} < 12.7 \text{ cm}$) and the rarity of large individuals, even in extensive forest inventories (Canham and Murphy 2017). Despite the absence of explicit size dependence in survival, indirect size effects are partially captured through asymmetric competition, where smaller individuals experience stronger competitive pressure. Another limitation shared with many forest-inventory-based models (Kunstler et al. 2021, Le Squin et al. 2021, Guyennon et al. 2023) is the focus on adult trees, even though fecundity can be influenced by climate (Clark et al. 2021), and the dynamics of recruitment may not necessarily align with those of adults (Serra-Diaz et al. 2016, Wason and Dovciak 2017, but see Canham and Murphy 2016).

The modular design of our framework enables direct integration of additional species and environmental predictors. For instance, additional covariates such as water balance or evapotranspiration could be incorporated to assess drought-induced mortality (Peng et al. 2011). Exploring interactions among climate, competition, and individual size may also improve predictions of demographic rates (Peng et al. 2011, Rollinson et al. 2016, Ford et al. 2017, Le Squin et al. 2021). A promising but computationally intensive extension would involve jointly fitting growth, survival, and recruitment models to explicitly capture their interdependence (Pang et al. 2024). Such an approach would allow the incorporation of ecological constraints, such as life-history trade-offs, by sharing information across demographic processes with abundant data (e.g. growth) and those with scarce data (e.g. recruitment). Understanding the ecological drivers underlying variation captured by random effects remains a priority. While our framework accounts for individual and plot-level uncertainty, additional attention to temporal variability in climate and competition will be important. Incorporating temporal stochasticity will improve predictions of species performance under changing environmental conditions and deepen understanding of population responses across space and time (Holt et al. 2022).

5 Conclusions

By integrating species-specific growth, survival, and recruitment models into Integral Projection Models, we showed that the population growth rate of 31 eastern North American tree species is consistently more sensitive to mean annual temperature than to competition, with sensitivity to precipitation substantially weaker. Demographic sensitivity varied systematically across thermal range positions: the relative dominance of climate over competition increased toward both cold and hot range limits, with these sensitivity gradients emerging along the continental thermal gradient shared across species rather than within each species' own range. This result suggests that range-edge demographic responses may emerge as a community-level phenomenon, and that increasing demographic sensitivity toward thermal extremes may provide a mechanistic pathway underlying the expected performance declines predicted by range-limit theories. Most of the variation in λ , however, remained tied to local plot conditions captured by random effects, indicating that fine-scale drivers not represented by our climate and competition covariates exert a stronger influence on demography than the broad-scale predictors typically used in range models. Together, these results provide a mechanistic basis for predicting how forest populations may reorganize under climate change, and a framework for understanding the demographic underpinnings of range-limit dynamics.

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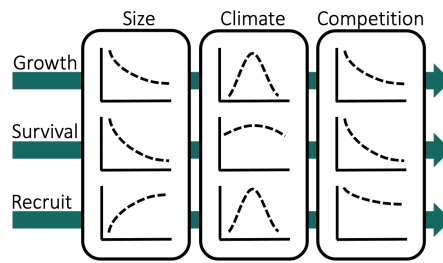
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719		across species' thermal ranges. Negative values indicate greater sensitivity to	
720		climate than to competition. Range positions were defined using the median mean	
721		annual temperature across all plots belonging to each thermal class. The larger	
722		and small bars represent the 20 and 70th quantile probabilities, respectively. . .	35

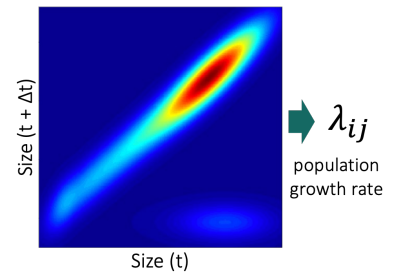
1) Forest inventory data



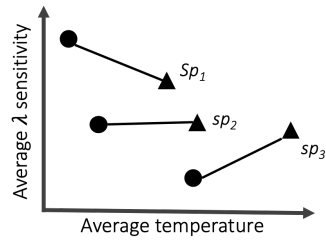
2) Demographic models



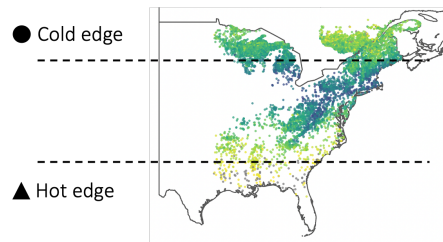
3) Integral Projection Model



6) Range-edge sensitivity



5) Spatial variation in λ sensitivity



4) Context-dependent λ sensitivity

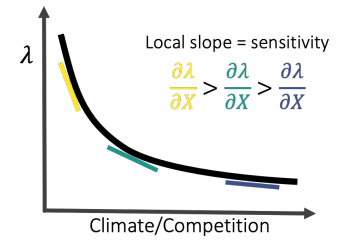


Figure 1

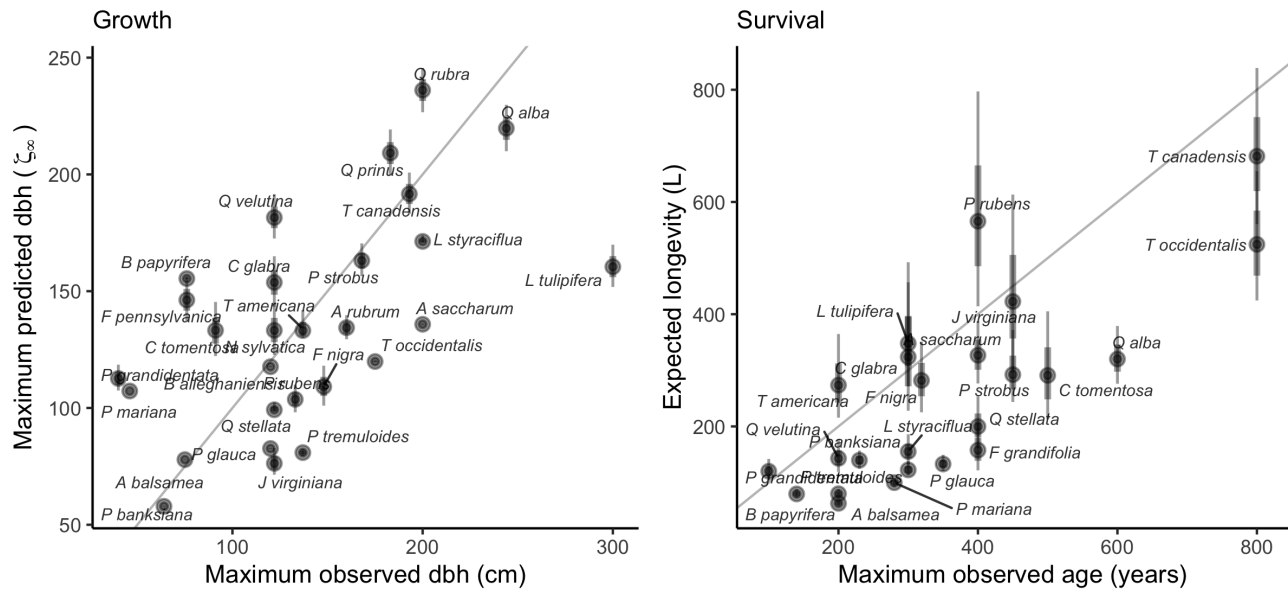


Figure 2

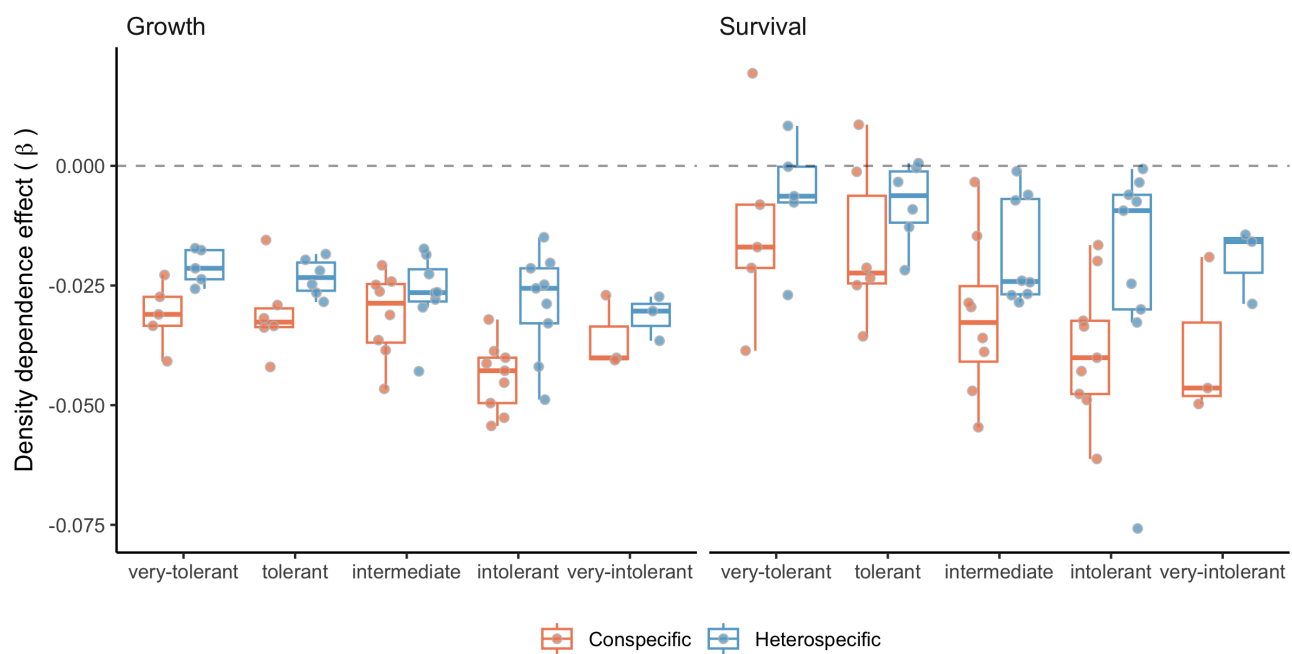


Figure 3

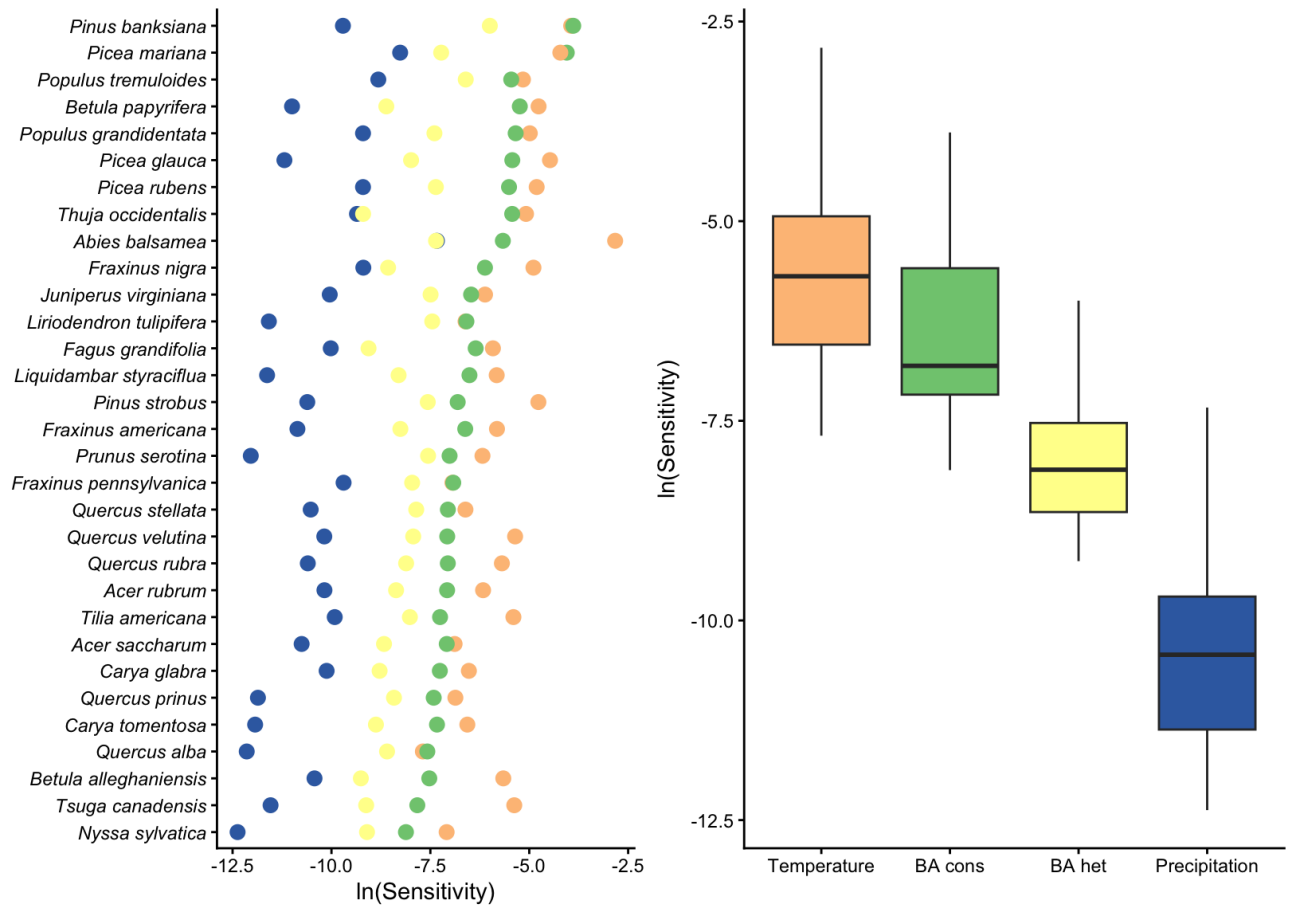


Figure 4

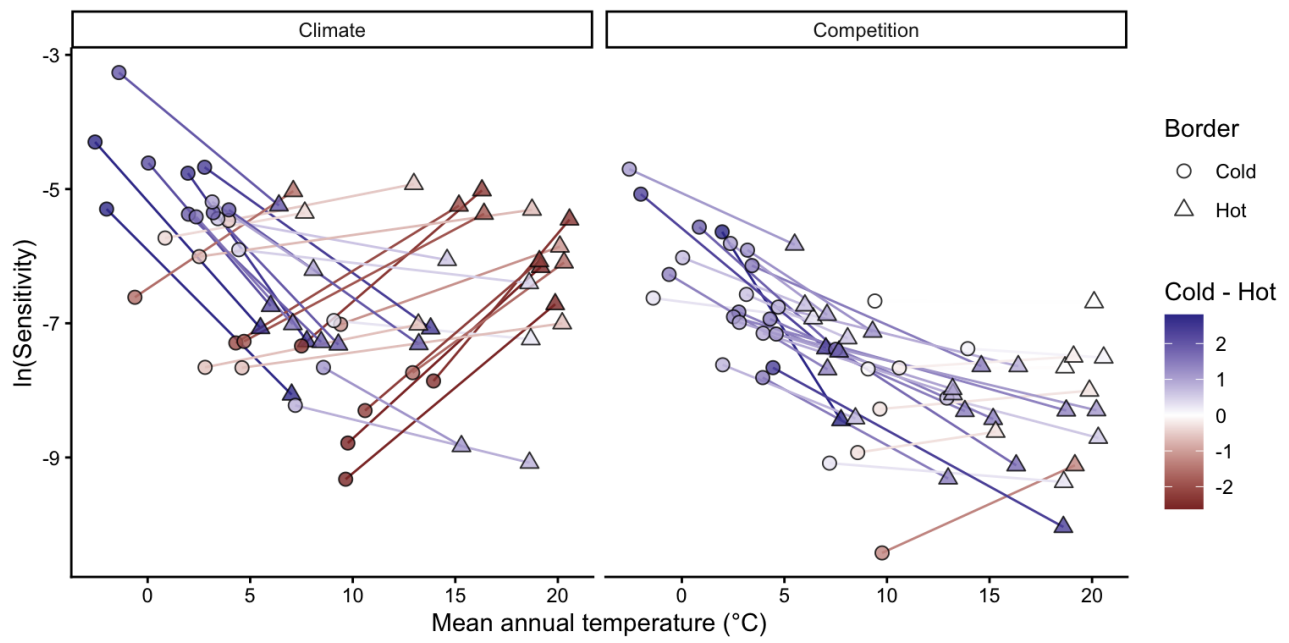


Figure 5

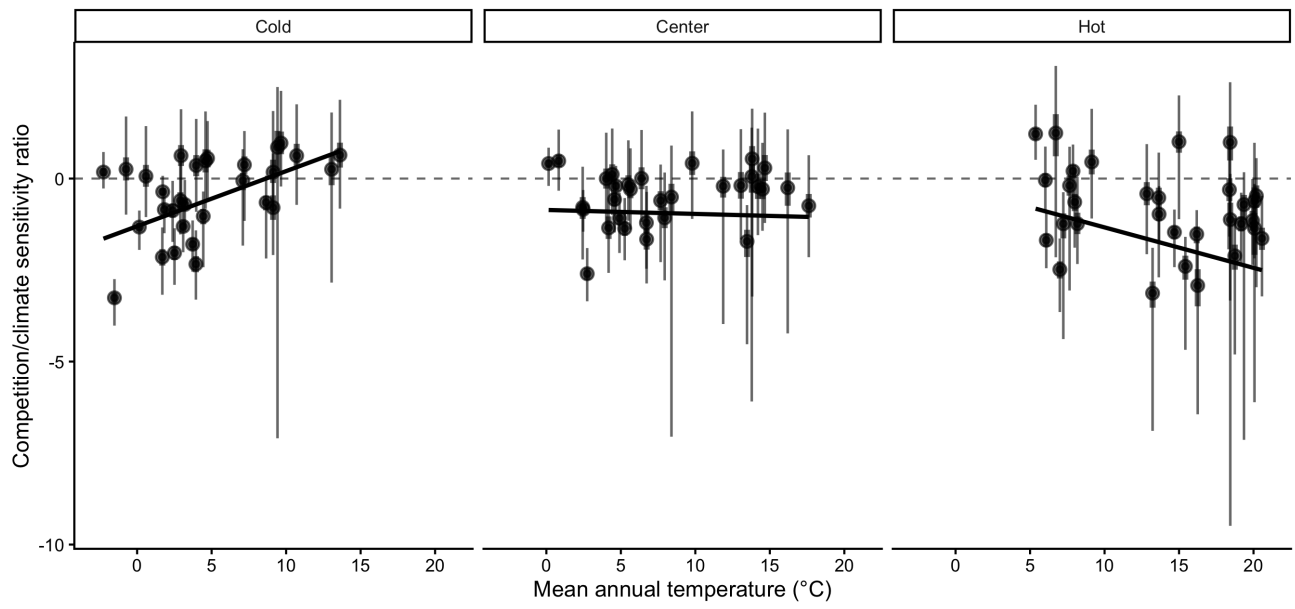


Figure 6